Module 11

Supporting Network Security Design

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Learning Objectives

- Explain the importance of network segmentation and use of trusted and untrusted zones.
- Describe security implications of internet of things (IoT) and industrial internet of things (IIoT).
- Explain the importance of physical security.

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Lesson 11.1

Zone-Based Security

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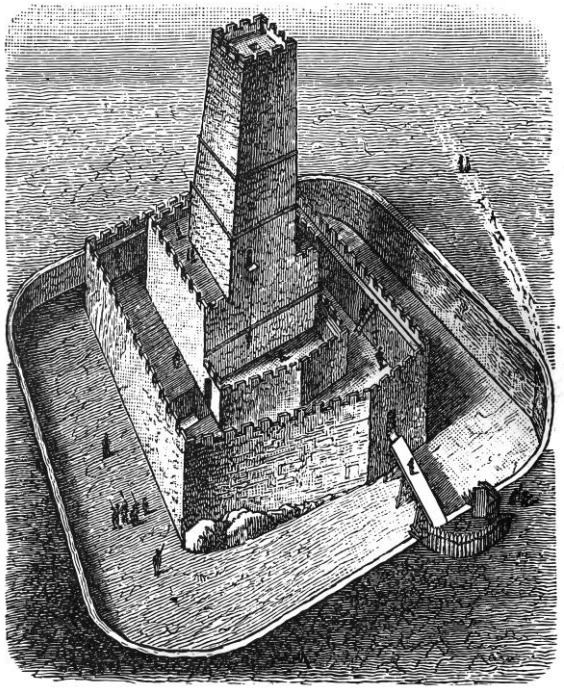
Zone-Based Security

Definition

- Network security model
- Organizes network into security zones
- Zones represent a region with specific security requirements

Security Model

- Ensures that traffic flows securely between zones
- Controls access
- Minimizes risk



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Network Security Zones

Private Security Networks

- Contain sensitive resources like internal servers, databases, and critical systems
- Access is tightly controlled

Private Client Networks

- Reserved for internal users and devices
- Includes workstations, laptops, and other endpoints

Guest Network

- For external visitors or temporary access
- Limited privileges and isolated from critical resources

Public Server Network

- Houses publicly accessible servers (e.g., web servers)
- Separated from internal networks to prevent direct attacks

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Perimeter Networks

Definition

- A secure buffer zone between an organization's internal network and the external, untrusted Internet

Characteristics

- Controls and filters traffic between internal and external networks
- Houses publicly accessible servers (e.g., web, email)
- Security appliances (firewalls, intrusion detection systems) are deployed here
- Separates sensitive data and systems from the public-facing network
- Provides policy enforcement, inspection, and auditing

Location

- Between external (untrusted) networks (e.g., Internet) and internal networks

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Screened Subnet

A firewall architecture that defines three separate subnets:

External Router	Internal Router	Perimeter Network
Separates the external network from the perimeter network	Separates the perimeter network from the internal network	Hosts servers accessible from both internal and external networks

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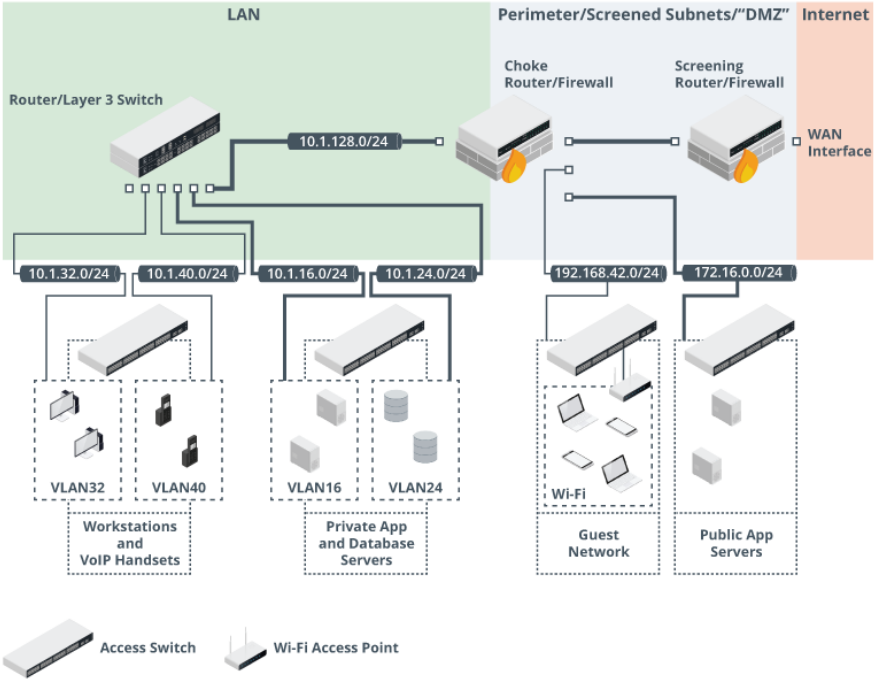
Screened Subnet Example



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Intrusion Detection and Prevention

IDS (Intrusion Detection System)

- Monitors network traffic, detects known attack patterns, and alerts administrators
- Traffic keeps flowing even during an attack
- Useful for quick setup but may delay response time

IPS (Intrusion Prevention System)

- Stops attacks immediately upon detection
- Traffic halts until investigation and decision
- Prioritizes prevention over alerting
- Ideal for critical assets but can impact legitimate traffic

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Activity: Discussion

- What is the primary difference between an IDS and an IPS?
- In a screened subnet architecture, what role does the external router play?

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Internet of Things

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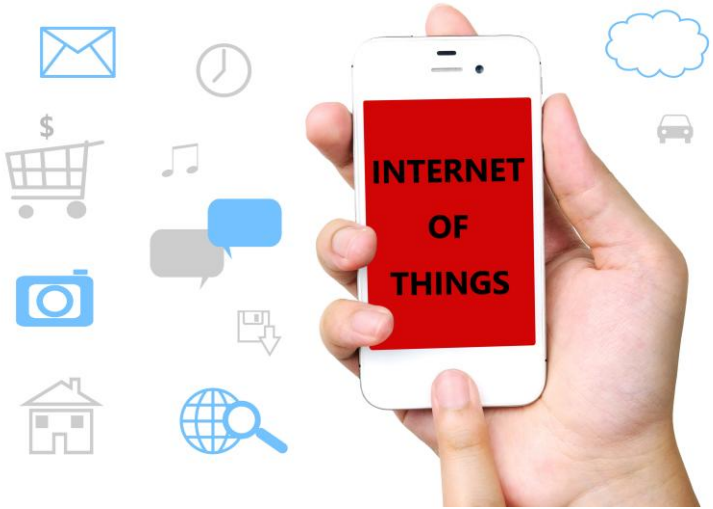
Think About It: IoT

What is the internet of things (IoT)?



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IoT Devices

Consumer devices

- Designed for personal use and convenience
- Often controlled through mobile devices

Industrial devices

- Monitor and control various aspects of industrial operations
- Improve efficiency and safety

Healthcare devices

- Support healthcare providers
- Help provide personalized, timely patient care

Agricultural devices

- Enhance precision farming
- Provide real-time data to optimize crop yield and livestock health

Smart city devices

- Help manage public resources efficiently
- Improve the quality of urban life

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Industrial Embedded Systems in IoT

Definition

- Specialized computing systems implanted in larger devices/systems
- Performs dedicated functions/tasks within industrial applications

Industrial Control Systems (ICS)

- Designed to manage and control industrial processes
- Used for automating plant operations

Supervisory Control and Data Acquisition (SCADA)

- Focuses on real-time data acquisition and control
- Monitors and controls equipment and conditions

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IoT Networking

IoT Network Integration

- Use unique identifiers for Internet connectivity, sometimes via intermediaries
- Standard networking products (Ethernet, Wi-Fi) might not always suit IoT device needs

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IoT Networking Standards

Operational Technology (OT) Networks

- Industrial settings, optimized for real-time data transfer
- Uses serial protocols or industrial Ethernet for deterministic transfers

Cellular Networks for IoT

- Narrowband-IoT (NB-IoT)
  - Low-power LTE variant, suitable for sending small data packets in challenging environments
- LTE Machine Type Communication (LTE-M)
  - Supports higher bandwidth (up to 1 Mbps) for more data-intensive IoT applications

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IoT Network Security

Definition & Importance

Protecting the network connecting IoT devices to back-end systems on the internet

Main Challenges

Increase the attack surface

Devices with weak security

Mitigation Strategies

Segmentation

Regular audits

Employee Training

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Managing IoT Security Risks

Consumer-Grade Devices

Risk: Vulnerabilities (weak default settings/inadequate patch management)

Management: Secure and patch IoT administrative interfaces

Smart Buildings

Risk: Compromise of entry mechanisms and climate/lighting control components

Approach: Configuration management and isolated network segments

ICS/SCADA

Challenges: Vulnerability (linked OT/IT networks) exploited by threat actors

Solution: Isolate management traffic and monitor cross-network links (access controls)

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Activity: True or False

IoT devices help to decrease the attack surface of a network.

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Lesson 11.3

Physical Security

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Physical Security

Why Physical Security is Important	Risks of Inadequate Physical Security	Mitigation Strategies
• First line of defense in protecting network resources • Can lead to data breaches, hardware theft, and vandalism	• Data Theft • Hardware Damage • Unauthorized Access	• Access Control Systems • Surveillance Systems • Environmental Controls • Regular Audits

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Locks and Secure Entry Systems

Locks and Secure Entry Systems

- Physical barriers and locks against unauthorized access

Locks and Badge Readers

- **Electronic Locks:** Secures entries electronically
- **Badge Reader:** Identifies and authorizes access via smart badges
- **Biometric Locks:** Authenticates using physical features

Rack System Locks

- **Secure Cabinets and Enclosures:** Protects IT assets from unauthorized access
- **Colocation Cages:** Provides separate access in shared data centers

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Camera-Based Surveillance Systems

Detection-Based Controls

- Act as a crucial layer of defense
- Ensure resilient security perimeters

Surveillance Applications

- Monitoring both perimeter areas and internal security zones
- Manned by security personnel or automated through video surveillance

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Camera Types

Fixed Cameras

- Recording all individuals passing through specific entry points
- Examples: access control vestibules

PTZ (Pan-Tilt-Zoom) Cameras

- Versatility for extensive area coverage
- Ability to zoom in on details and track movements within large spaces

CCTV Network Integration

- Cameras typically connect via coaxial cable to a multiplexer
- Enable image display across multiple screens
- Include both control and recording functionalities
- Integration into IP networks for enhanced flexibility and efficiency
- Smaller IP cameras support Power over Ethernet (PoE)

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Geofencing Overview

Geolocation and its Uses

- Identifies the physical position of a device using network attributes
- Location services for determining a device's current position

Location Services

- GPS (Global Positioning System)
- IPS (Indoor Positioning System)

Geofencing Explained

- A virtual boundary created in real-world geography
- Controlling device functionalities (camera and video)
- Context-aware authentication

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Geofencing Overview

Applications in Organizations

- Creation of virtual perimeters around specific areas (e.g., office properties)
- Restricts device functions (e.g., camera use outside the designated area)

How Does it Work?

- Device position identified through location services
- Enforcement and policy management facilitated by Mobile Device Management (MDM) software

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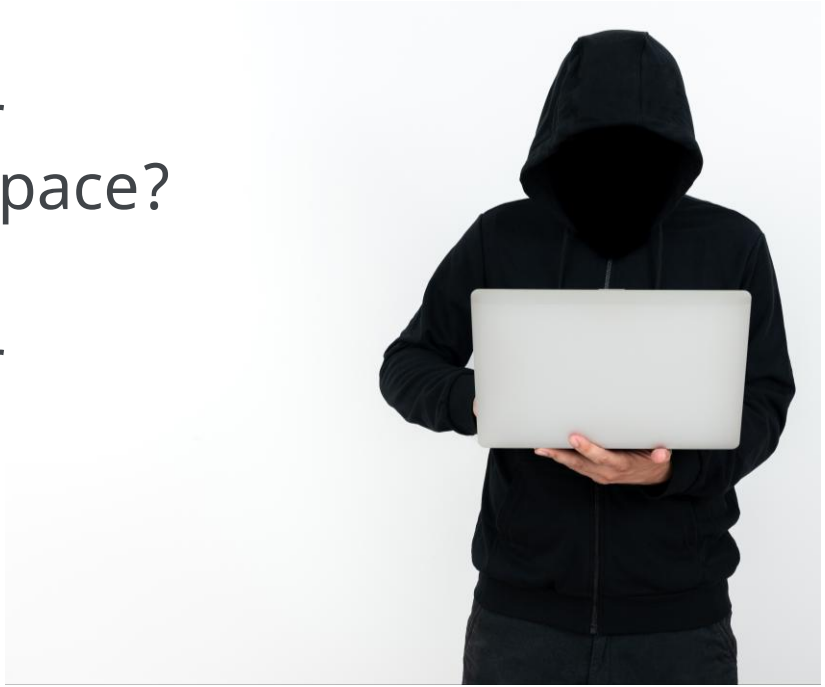
Activity: Worst Case Scenario

- What if a would-be hacker accessed a shared office space?
- What if a would-be hacker accessed a server rack?



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Summary

- **Perimeter Defense**: Utilize firewalls/proxy servers in screened subnets
- **Internal Zone Protection**: Implement firewalls for internal zones/hosts
- **Intrusion Detection/Prevention**: Deploy IDS/IPS behind firewalls
- **IoT Management**: Develop a strategy for segmented network zones
- **Site Access Control**: Assess and implement access control measures

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